

PROBLEM SET 3—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions: Due on 16/10/19 at the beginning of the lecture. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission. Don't forget to attach the cover sheet.

- Let the agent be a drinker who plans to buy at most one bottle of wine within a month. His utility is $U = \theta q - t$ where q is the quality he buys and θ is a positive parameter that indexes his taste for quality. If he decides not to buy any wine, his utility is just 0. Suppose that there are n types of drinkers and that $\theta_1 < \dots < \theta_n$. Their respective prior probabilities are $\pi_1, \dots, \pi_n$ with $\sum_{i=1}^n \pi_i = 1$. The principal is a local monopoly on the wine market. He can produce wine with any quality $q \geq 0$; the production of a bottle of wine with quality q costs him $c(q) = \frac{q^2}{2}$.

- Show that the only binding constraints are the downward adjacent incentive constraints, i.e.

$$\theta_i q_i - t_i \geq \theta_i q_{i-1} - t_{i-1} \quad i = 2, \dots, n$$

and the participation constraint for the lowest type, i.e.

$$\theta_1 q_1 - t_1 \geq 0$$

- Characterize as far as you can the solution to the monopolist's problem.

For the remainder of the problem, assume that $n = 3$ and that $\theta_1 = 1, \theta_2 = 5, \theta_3 = 9$.

- Suppose that $\pi_1 = 0.85, \pi_2 = 0.08$ and that $\pi_3 = 0.07$. Compute the optimal qualities for each consumer, the optimal transfers and the monopolist's profits.
- Suppose now that $\pi_1 = \pi_2 = \pi_3 = 1/3$. Compute again the optimal qualities, transfers and the monopolist's profits.
- Compare the results from (c) and (d). What did you find? In which of the two cases is the monopolist performing better? Can you provide an economic interpretation of your results?

Solution

- We start with an approach similar to the one in the lecture slides. The monopolist's problem is

$$\max_{\{(q_i, t_i)\}_{i=1, \dots, n}} \sum_{i=1}^n \pi_i \left(t_i - \frac{q_i^2}{2} \right) \quad (1)$$

subject to

$$\theta_i q_i - t_i \geq \theta_i q_j - t_j, \quad \forall i, j \quad (2)$$

$$\theta_i q_i - t_i \geq 0, \quad \forall i \quad (3)$$

$$q_i \geq 0, \quad \forall i \quad (4)$$

The previous problem is equivalent to the relaxed problem:

$$\max_{\{(q_i, t_i)\}_{i=1, \dots, n}} \sum_{i=1}^n \pi_i \left(t_i - \frac{q_i^2}{2} \right) \quad (5)$$

subject to

$$\theta_i q_i - t_i \geq \theta_i q_{i-1} - t_{i-1}, \quad \forall i = 2, \dots, n \quad (6)$$

$$\theta_1 q_1 - t_1 \geq 0, \quad (7)$$

$$q_i \geq q_{i-1}, \quad \forall i = 2, \dots, n \quad (8)$$

$$q_i \geq 0, \quad \forall i \quad (9)$$

First we will show that when monotonicity (8) holds, then all the incentive constraints (2) hold if and only if the downward constraints (6) and the upward constraints

$$\theta_i q_i - t_i \geq \theta_i q_{i+1} - t_{i+1}, \quad \forall i = 1, \dots, n-1, \quad (10)$$

hold. Clearly the full set of incentive constraints (2) imply both the downward (6) and upward (10) constraints. To prove the other direction, fix an index $i \in \{2, \dots, n\}$. Now we need to pick an arbitrary index j but as mentioned in class, we have two cases:

- i. $j < i$: This case was done in class and I am going to skip it. Follow the lecture notes to prove this case.
- ii. $j > i$: This is the case that I asked you to verify in the lecture notes. I am going to prove this case here.

Thus, take any $j > i$. Notice that

$$\theta_i q_i - q_j \theta_i = \sum_{k=1}^{j-1} \theta_i q_k - \theta_i q_{k+1} \geq \sum_{k=1}^{j-1} \theta_k q_k - \theta_k q_{k+1} \geq \sum_{k=1}^{j-1} t_k - t_{k+1} = t_i - t_j,$$

where the first equality is by a telescopic sum argument, the second inequality is by increasing differences, the third inequality is by upward incentive compatibility (10) and the fourth equality is again by a telescopic sum argument.

We continue with the proof. We proceed in 4 steps:

Step 1 The constraint set of P is a subset of RP. Notice that the full set of incentive compatibility restrictions (2) imply that

$$\theta_{i-1} q_i - \theta_{i-1} q_{i-1} \leq t_i - t_{i-1} \leq \theta_i q_i - \theta_i q_{i-1}$$

which implies $(q_i - q_{i-1})(\theta_{i-1} - \theta_i) \geq 0$. Since types are such that $\theta_i \geq \theta_{i-1}$ (this is the SCP in for this model), then we have that the previous inequality implies that $q_i \geq q_{i-1}$.

Step 2 At the solution to RP, (6) holds as an equality for all i and (7) also holds as an equality. Otherwise, we can increase all t_j , $j \geq i$ or t_1 by some $\epsilon > 0$, increasing the profit of the monopolist.

Step 3 Solutions to RP satisfy the constraints of P. Since (6) hold with equality and $q_i \geq q_{i-1}$, we have that

$$t_i - t_{i-1} = \theta_i q_i - \theta_i q_{i-1} \geq \theta_{i-1} q_i - \theta_{i-1} q_{i-1},$$

that is, all upward IC constraints (10) hold, and by the first result we proved here, this implies that the full set of IC constraints (2) hold.

Step 4 The participation constraints of the lowest type (7) implies all participation constraints (3). Note that the full set of IC (2), PC for the lowest type (7), and single crossing ($\theta_i \geq \theta_{i-1}$) imply that

$$\theta_i q_i - t_i \geq \theta_i q_1 - t_1 \geq \theta_1 q_1 - t_1 = 0.$$

Thus we have shown that at an optimal solution, the only relevant constraints are (6) and (7) and that these need to hold with equality, thus completing the proof.

- (b) We can follow the general development in the slides to characterize the solution. Foregoing the monotonicity constraint in the relaxed problem, we get:

$$q_n = \theta_n, \quad (11)$$

$$q_i = \max \left\{ \theta_i - \frac{1 - \Pi_i}{\pi_i} (\theta_{i+1} - \theta_i) \right\}, \quad (12)$$

where $\Pi_i = \sum_{k=1}^i \pi_k$. We can see from (11) that there is no distortion for the top type, and from (12) there will be a distortion. Furthermore, the expression in (12) takes into consideration the possibility of *corner solutions*, i.e. when the first order condition identifies a negative solution for i , then we put $q_i = 0$.

Note that in order for the solution to be monotonic, we would have to impose more restrictions on the parameters. Let $\Delta_i = \theta_{i+1} - \theta_i$ for every $i = 1, \dots, n-1$, and let $H_i = \frac{1 - \Pi_i}{\pi_i}$ for $i = 1, \dots, n$

- A *necessary and sufficient* condition for monotonicity is: $\frac{\Delta_i}{\Delta_{i+1}} \geq \frac{H_{i+1}}{1 + H_i}$ for $i = 1, \dots, n-1$.
- *Sufficient conditions* for monotonicity are that H_i should be decreasing with i and that also Δ_i should be decreasing with i . These two conditions are the ones that I mentioned in the last slide of the lecture: the inverse hazard rate should be decreasing in i (monotone hazard rate property) and that the difference in marginal values between types adjacent types should also be decreasing in i .

To characterize the transfers, we use the binding constraints (7) and (6):

$$t_1 = \theta_1 q_1, \quad (13)$$

$$t_i = \theta_i (q_i - q_{i-1}) + t_{i-1}, \quad i = 2, \dots, n. \quad (14)$$

- (c) Quantities, transfers and profits are:

$$(q_1, q_2, q_3) = \left(\frac{5}{17}, \frac{3}{2}, 9 \right), \quad (15)$$

$$(t_1, t_2, t_3) = \left(\frac{5}{17}, \frac{215}{34}, \frac{1255}{17} \right), \quad (16)$$

$$\text{Profits} = \frac{1007}{340}. \quad (17)$$

- (d) Quantities, transfers and profits are:

$$(q_1, q_2, q_3) = (0, 1, 9), \quad (18)$$

$$(t_1, t_2, t_3) = (0, 5, 77), \quad (19)$$

$$\text{Profits} = \frac{41}{3}. \quad (20)$$

- (e) The monopolist performs better under (d) than in (c). Notice that the monopolist is shutting down consumption for the lowest type under scheme (d). Thus, at the cost of not contracting with the lowest type, the monopolist is able to extract more rent from higher types, and this trade-off is optimal.

Define the *cumulative distribution functions* for each case as

$$G(\theta) = \begin{cases} 0 & \theta < \theta_1 \\ 0.85 & \theta_1 \leq \theta < \theta_2 \\ 0.93 & \theta_2 \leq \theta < \theta_3 \\ 1 & \theta_3 \leq \theta \end{cases} \quad F(\theta) = \begin{cases} 0 & \theta < \theta_1 \\ 1/3 & \theta_1 \leq \theta < \theta_2 \\ 2/3 & \theta_2 \leq \theta < \theta_3 \\ 1 & \theta_3 \leq \theta \end{cases}$$

Note that $F(\theta) \leq G(\theta)$ for every θ . Intuitively, this means that F assigns higher probabilities to higher types than G . This means that there is a higher chance of meeting a higher type than a lower type from F than from G . Because of this, the monopolist is confident that by shutting down contracting with the lowest type under F is profitable.